

TrustLockAI Architecture Deep-Dive

How a sealed, air-gapped AI server brings modern language models to adverse-event case work — with the model weights and your patient data never leaving your network. This document is written for safety, IT security, and validation stakeholders evaluating TrustLockAI against GVP, GxP, 21 CFR Part 11, and DPDP obligations.

Why on-premise is the only viable path

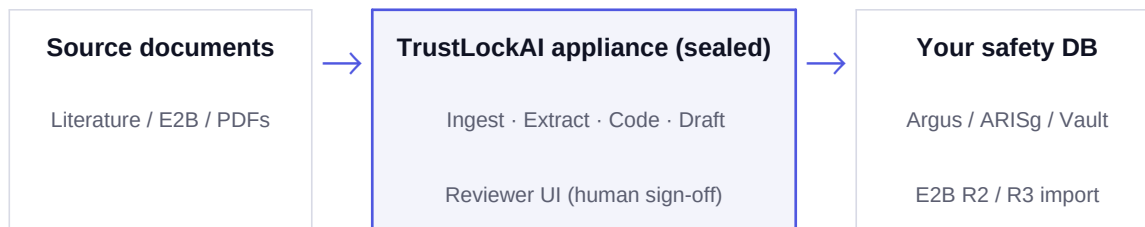
Case volumes and literature surveillance keep rising while trained safety associates stay scarce. Language models can absorb the repetitive intake, extraction, and coding work — but individual case safety reports contain special-category personal data that cannot be sent to a public cloud model under GVP Module VI, GxP, and India's DPDP Act. The result: the workflow that would benefit most from AI stays manual. TrustLockAI resolves the conflict by moving the model to the data, inside your own four walls.

The appliance at a glance

Form factor	Sealed rack-mounted server (or hardened VM image for your private VPC).
Model	Ships inside the appliance. No model API calls leave the box — ever.
Network posture	Zero third-party egress by architecture; demonstrable with the network cable pulled.
Integration	Standard APIs into Argus, ARISg, or Veeva Vault Safety; returns E2B (R2/R3).
Human control	Assistive only — associates review, edit, and sign off on every output.
Audit	Append-only, versioned, explainable steps for Part 11 / GAMP 5 / GVP scrutiny.
Commercial	Delivered as a server you own — not a per-token cloud subscription.

Reference architecture

TrustLockAI sits beside your existing safety database as an isolated processing node. Documents flow in, structured E2B flows out, and a reviewer UI keeps a human in the loop. Nothing transits a public network.



Processing pipeline

- **Ingest** — literature articles, source PDFs, and inbound E2B are queued locally.
- **Extract** — the on-box model pulls patient, drug, event, and seriousness fields.
- **Code** — MedDRA coding assist proposes LLTs/PTs with traceable rationale.
- **Draft** — a case narrative is generated from the extracted, reviewer-confirmed fields.
- **Review** — associates verify and sign off; corrections are captured for audit.
- **Return** — results are written back as E2B into your safety database.

Data-flow & isolation controls

- All inference runs on local accelerators inside the appliance; no token, prompt, or document body is transmitted to any external endpoint.
- Outbound network can be physically or logically severed without degrading core function — the litmus test we run live during the air-gapped demo.
- Model weights are stored on the appliance; there is no phone-home, telemetry, or model-update channel unless you explicitly open and approve one.
- Identity, access, and key management bind to your existing directory and HSM where available.

Compliance mapping

TrustLockAI is designed so that the AND of “use modern AI” and “keep data resident” is satisfiable. The table maps common obligations to architectural controls.

Obligation	How TrustLockAI addresses it
DPDP Act — data residency	Patient data never leaves your network boundary; processing is on-box.
GVP Module VI — case integrity	Deterministic, logged transformations; human verification before submission.
21 CFR Part 11 — e-records	Append-only audit trail, versioning, attributable sign-off, time-stamping.
GAMP 5 — CSV	Documented intended use, risk-based test evidence, change control.
Human oversight	Assistive-only; the system never auto-submits a case without sign-off.

Validation & deployment

A typical engagement runs as a fixed-scope pilot on your published-literature cases, followed by validation against your SOPs and a production cut-over.

1. Air-gapped demo	Extraction, coding, narrative, and E2B export shown with the network cable pulled.
2. Pilot	Fixed-scope run on your public-data cases; accuracy and throughput baselined.
3. Validation	IQ/OQ/PQ against your SOPs; CSV package assembled with your QA.
4. Production	Appliance racked in your network/VPC; integrated to your safety DB.

What it handles today

- Literature case intake & screening
- Case extraction (patient · drug · event · seriousness)
- MedDRA coding assist
- Case narrative drafting
- E2B output to your safety database

Next step: book the air-gapped demo and a free data-leak & cloud-cost audit of your case-intake workflow at www.nfinityinfotech.com. The demo runs on public data — no patient data required.